Hyperspectral Anomaly Detection Based on Fractional Domain Sensing and Contrastive Statistical Regularized Network

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***Abstract*—Hyperspectral anomaly detection (HAD) is a highly challenging task, primarily because weak anomalous targets within high-dimensional spectral data are easily obscured by complex and heterogeneous backgrounds. However, existing methods still have limitations. They typically rely on strong background assumptions or focus solely on spectral or spatial features, neglecting frequency-domain information. This makes it difficult to effectively distinguish subtle anomalies in complex scenes. Therefore, this study proposes a method based on fractional domain sensing and contrastive statistical regularized network (FDCRN), which adopts the diffusion model as the basic framework for background suppression and optimizes anomaly detection performance from three dimensions, including frequency-domain enhancement, spatiospectral fusion, and distribution correction. First, we design a lightweight fractional Fourier transform (LFrFT) module, which adaptively modulates the spectral frequency components through learnable fractional-order parameters and combines frequency-domain features to enhance the capture of subtle differences of weak and small targets. Second, we propose a multiscale spatial–spectral enhancement (MSSE) module, which extracts multiscale spatial information by using depth-separable convolution and fuses spectral features to enhance the spatial–spectral correlation of weak and small targets. Third, we introduce a statistical adaptive and contrastive learning (SACL) module to dynamically correct the feature distribution shift and increase the semantic distance between the background and the anomaly. Experiments on multiple real hyperspectral datasets show that FDCRN outperforms existing methods in detection accuracy, response intensity to weak and small targets, and adaptability to complex scenes, providing a more efficient solution for HAD in complex scenarios.**

***Index Terms*—Contrastive learning, diffusion model, hyperspectral anomaly detection (HAD), lightweight fractional Fourier transform (LFrFT), multiscale spatial–spectral.**

Received 23 February 2026; revised 2 July 2026; accepted 17 August 2026. Date of publication 19 September 2026; date of current version 27 October 2026. This work was supported in part by the National Natural Science Foundation of China (NSFC) under Grant NSFC 62401062, Grant 62222102, and Grant 62171023; in part by the National Key Research and Development Program of China under Grant 2024YFB3909800; in part by the Open Foundation of Hubei Key Laboratory of Applied Mathematics (Hubei University) under Grant HBAM202301; and in part by the Fundamental Research Funds for the Central Universities under Grant FRF-TP-25-038. *(Corresponding author: Junjun Yin.)*

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Digital Object Identifier 10.1109/TGRS.2026.3735615

I. Introduction

HYPERSPECTRAL remote sensing technology plays an important role in practical applications such as ecological environment monitoring [1], [2], [3], [4], geological disaster early warning [5], [6], [7], [8], and military target identification [9], [10], [11], [12]. Hyperspectral image (HSI) classification, covering cross-domain adaptive learning and open-set recognition paradigms, constitutes a fundamental research branch for land-cover mapping and semantic interpretation [13], [14], [15]. Hyperspectral unmixing is devoted to decomposing mixed pixels and addressing endmember spectral variability to mitigate the negative influence of mixed pixels prevalent in real scenes [16], [17]. Hyperspectral change detection focuses on capturing subtle temporal spectral variations for monitoring land cover evolution across multitemporal observations [18], [19]. Among these, hyperspectral anomaly detection (HAD) is one of the core technologies supporting remote sensing-based Earth observation and resource exploration [20], [21], [22], [23]. HSIs capture information about ground objects across dozens to hundreds of continuous spectral bands. Each pixel corresponds to a unique spectral curve [24], [25], which precisely represents the material composition and physical properties of the objects. This ability enables the detection of subtle differences between ground objects that are often difficult to discern with traditional imaging methods [26], [27]. However, HSIs face several challenges, including high dimensionality and abundant redundant information [28], [29], as well as complex surface coverage, mixed pixels from multiple ground objects, and limited spatial resolution [30], [31], [32]. These characteristics make it difficult for conventional detection methods to be directly applied to hyperspectral data. Therefore, developing dedicated anomaly detection technologies tailored to the unique characteristics of HSIs is crucial for improving the efficiency of remote sensing data utilization and expanding its practical applications.

In its early stages, HAD is primarily focused on statistical modeling and representation learning. A classic benchmark in

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this area is the Reed–Xiaoli (RX) algorithm [33]. However, RX's reliance on global statistics makes it less effective in scenarios with uneven backgrounds, as it assumes a Gaussian distribution that often deviates from the actual distribution of ground objects. To address these limitations, researchers have proposed several improvements. Local window algorithms, such as local adaptive RX (LAIRX), focus on local background statistics to mitigate distribution heterogeneity [34]. The kernelized RX (KRX) methods [35] tackle nonlinear relationships through kernel mapping. The uniform target detection RX (UTDRX) method replaces the matching signal with a uniform vector, enhancing target detection performance [36]. While RX and its variants [37], [38], [39], [40] all center around statistical modeling of background distributions, they struggle to adapt to the nonlinear and heterogeneous distributions commonly encountered in real hyperspectral imagery.

In traditional methods based on representation learning, low-rank sparse matrix factorization (LSMAD) [41] relies solely on the low-rank and sparse characteristics of the spatial domain for modeling and fails to utilize frequency-domain features to enhance the spectral discrimination of weak and small targets. In contrast, the kernel isolation forest detection (KIFD) method [42] isolates anomalous targets but lacks adaptive modeling of the frequency-domain structure. Moreover, some spatial information is iteratively optimized through local windows without forming a multiscale spatiospectral joint enhancement mechanism, leading to insufficient capture of the spatial correlation of weak and small targets. These methods do not sufficiently leverage frequency-domain structural information, in which subtle target–background discrepancies are embedded, leading to reduced sensitivity to high-frequency perturbations and limited capability in representing weak and small targets.

In recent years, deep learning has emerged as a research hotspot due to its strong feature extraction capability and powerful nonlinear modeling ability. Data-driven methods based on architectures such as autoencoders (AEs) [43], [44], [45], [46] and generative adversarial networks (GANs) [47], [48], [49], [50] have significantly improved background suppression and anomaly discrimination in complex scenes by learning deep latent representations or exploiting reconstruction errors. In addition, the low-rank regularization and total variation anomaly detection (LARTVAD) method [51] preserves spatial and spectral structures through tensor modeling and incorporates total variation regularization to enhance structural consistency. Despite these advances, most deep learning-based HAD methods still operate at the pixel level or rely on relatively shallow spatial and spectral fusion. They usually lack explicit modeling of the frequency domain and, therefore, fail to exploit frequency structural differences that are critical for effective anomaly background discrimination. A recently proposed high–low-frequency feature-guided diffusion model (HFGDM) [52] alleviates this limitation by introducing explicit separation and guidance of high-/low-frequency components within the diffusion forward and reverse processes, yet it still suffers from insufficient interaction between multiscale spatial–spectral features.

Meanwhile, some approaches, particularly early purely spectral AE-based methods, do not sufficiently incorporate spatial contextual information and mainly depend on pixelwise spectral features. As a result, the spatial correlation of weak and small targets is easily submerged by background noise, leading to insufficient detection response strength and poor stability. Some recent methods, including Transformer-based AE framework (TAEF) [53], feature enhancement network via reverse distillation (FERD) [23], and multiscale dual-domain reconstruction masked network (MrsNet) [54], have achieved notable performance improvements. They do not fundamentally resolve the tradeoff between background modeling adaptability and anomaly feature discriminability. Consequently, achieving robust and reliable detection in complex real-world hyperspectral remote sensing scenarios remains a challenging problem.

The background suppression diffusion model (BSDM) [55] introduces diffusion modeling into HSI processing by treating complex backgrounds as stochastic noise, improving robustness in cluttered scenes. However, for challenging scenarios involving weak and small targets, diffusion-based modeling still faces several challenges. In particular, frequency-domain characteristics of hyperspectral signals are not sufficiently exploited, spatial–spectral contextual correlations are not fully utilized, and background distribution imbalance under complex environmental interference is not explicitly addressed. Motivated by these challenges, this study proposes a fractional domain sensing and contrastive statistical regularized network (FDCRN). To enhance frequency-domain discrimination, a lightweight fractional Fourier transform (LFrFT) module is introduced to adaptively modulate spectral frequency components. By integrating frequency-enhanced features with the original spectrum through a residual pathway, subtle differences in weak and small targets are effectively captured. A multiscale spatial–spectral enhancement module (MSSE) is designed to jointly exploit spatial information and spectral features. This module strengthens spatial consistency and spectral contrast of anomalous targets, leading to more stable and robust detection of weak targets. Furthermore, to address background distribution imbalance, a statistical adaptive module combined with contrastive learning (SACL) projection head is constructed. This mechanism dynamically adjusts feature representations and enhances the separability between background and anomalous features. The contributions of this study are given as follows.

1) FDCRN is proposed for anomaly detection, which jointly exploits frequency-domain enhancement, spatial–spectral feature interaction, and distribution regularization to enable more reliable discrimination of weak and small targets under complex background conditions.
2) LFrFT is proposed to adaptively modulate spectral frequency components and integrate frequency-enhanced features with the original spectrum, enabling effective capture of subtle frequency-domain differences of weak and small targets and improving anomaly separability under complex background conditions.
3) The MSSE module is designed to jointly exploit multiscale spatial information and spectral features,

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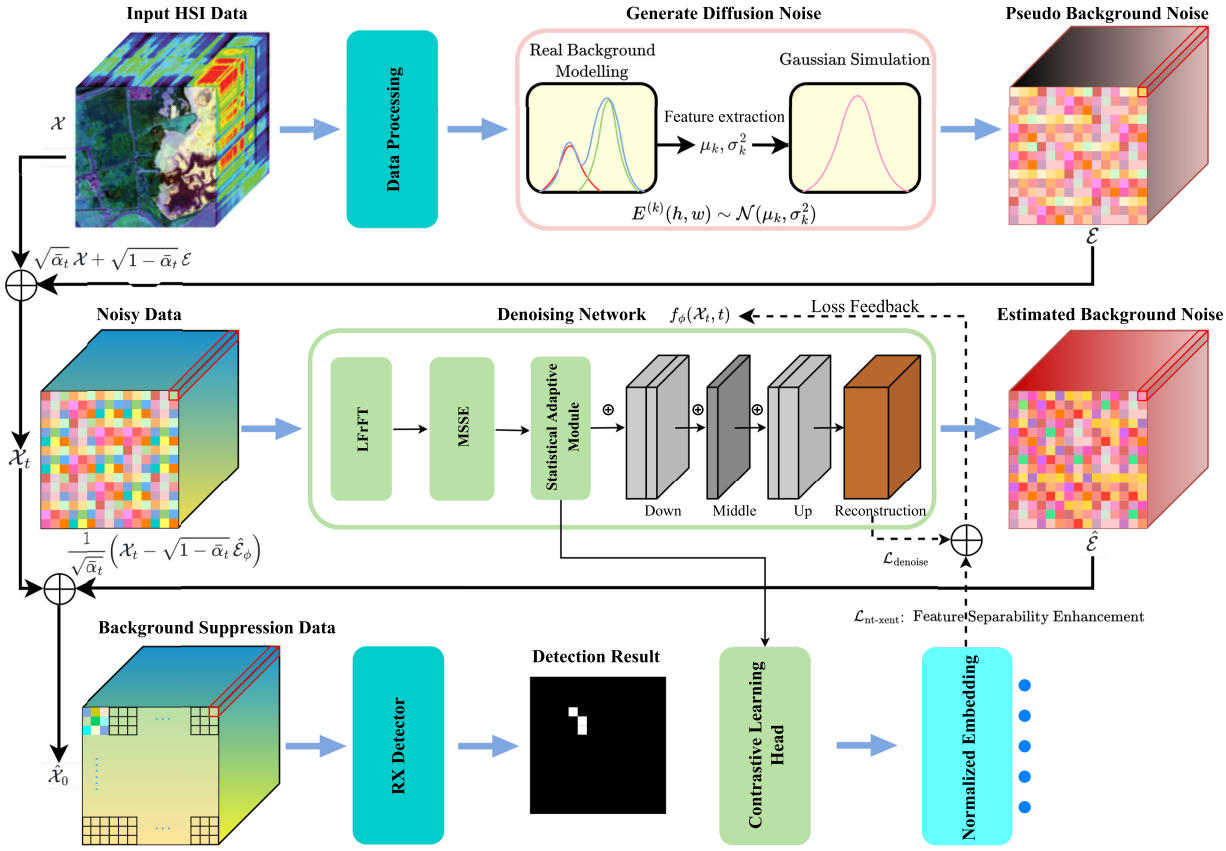


Fig. 1. Overall framework of the proposed FDCRN, which integrates three key modules. The LFrFT module adaptively regulates spectral frequency components via learnable fractional-order parameters to capture subtle variations of weak targets. The MSSE module extracts multiscale spatial information using depthwise separable convolutions and fuses spectral features, thereby enhancing sensitivity to weak targets. The SACL module dynamically corrects feature distribution shifts and improves the separability between background and anomaly features.

enhancing spatial consistency and spectral contrast of anomalous targets and enabling more stable and robust detection of weak targets in complex scenes.

4) The SACL mechanism is constructed to address background distribution imbalance by dynamically adjusting feature representations and enhancing semantic separability between background and anomalous features, thereby improving detection robustness.

II. Proposed Method

The overall framework of the proposed FDCRN is illustrated in Fig. 1. Given an input HSI, the proposed method is formulated within the deep background suppression framework (DBSF), which serves as the basic background suppression mechanism for modeling complex backgrounds. The overall processing pipeline consists of three main stages. First, the LFrFT module is incorporated to enhance spectral representations in the fractional frequency domain, thereby facilitating the discrimination of weak and small targets. Second, based on the enhanced spectral features, the MSSE module is employed to extract joint spatial–spectral information and promote the spatial consistency of potential anomalies. Third, the SACL mechanism is introduced to adaptively regulate feature distributions and generate the final anomaly map, which improves robustness in complex scenes. The detailed configurations of each component are presented in Sections II-B–II-D.

A. Deep Background Suppression Framework

For $\mathcal{X} \in \mathbb{R}^{H\times W\times B}$, here, H, W, and B denote the height, width, and number of spectral bands of the HSI, respectively. The pseudobackground Gaussian noise $\mathcal{E} \in \mathbb{R}^{H\times W\times B}$ consistent with background statistics is generated. For each spectral band $k \in \{1, 2, \ldots, B\}$, the mean μ_k and variance σ_k^2 are computed by averaging over all spatial pixels $(h, w) \in \{1, \ldots, H\} \times \{1, \ldots, W\}$ of the kth band matrix $X^{(k)} \in \mathbb{R}^{H\times W}$. Each pixel in the kth noise matrix $E^{(k)} \in \mathbb{R}^{H\times W}$ is independently sampled as

$$E^{(k)}(h,w) \sim \mathcal{N}\left(\mu_k, \sigma_k^2\right) \quad \forall\,(h,w) \tag{1}$$

where $\mathcal{N}(\mu_k, \sigma_k^2)$ denotes the Gaussian distribution with mean μ_k and variance σ_k^2. The full pseudobackground noise tensor $\mathcal{E}$ is formed by stacking $\{E^{(k)}\}_{k=1}^{B}$ along the spectral dimension.

Due to the nonlinear and highly complex distribution of hyperspectral backgrounds, injecting Gaussian noise in a single step cannot faithfully approximate its intrinsic statistical

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characteristics. Such abrupt perturbation would severely distort the background structure and potentially overwhelm weak anomalous signatures. Moreover, it contradicts the Markov chain formulation of diffusion models, which rely on progressive noise accumulation and gradual distribution smoothing. Therefore, we construct a forward diffusion process that progressively introduces pseudobackground noise to model the background distribution. Let the total diffusion steps be T and define the linear noise schedule as

$$\beta_s = \beta_{\text{start}} + \frac{s-1}{T-1}(\beta_{\text{end}} - \beta_{\text{start}}), \quad s = 1, \ldots, T \tag{2}$$

where $\beta_s \in (0, 1)$ denotes the noise injection rate at step s. The cumulative product is defined as

$$\bar{\alpha}_t = \prod_{s=1}^{t} (1 - \beta_s). \tag{3}$$

The noisy tensor at step t is

$$\mathcal{X}_t = \sqrt{\bar{\alpha}_t}\,\mathcal{X} + \sqrt{1 - \bar{\alpha}_t}\,\mathcal{E}. \tag{4}$$

The objective of the diffusion model is to progressively mix the original HSI with Gaussian noise and train a denoising network $f_\phi(\mathcal{X}_t, t)$ to predict the injected noise at different diffusion stages, where ϕ denotes the network parameters and t is the time-step embedding. Owing to its equivalence to Gaussian maximum likelihood estimation, the L_2 loss enables the network to accurately model the Gaussian noise distribution by minimizing the squared Euclidean distance between the predicted noise and the ground-truth noise. The training objective is formulated as

$$\mathcal{L}_{\text{denoise}} = \mathbb{E}_{\mathcal{X},\mathcal{E},t}\left[\|\mathcal{E} - f_\phi(\mathcal{X}_t, t)\|_2^2\right]. \tag{5}$$

During inference, the trained network predicts the noise $\hat{\mathcal{E}}_\phi = f_\phi(\mathcal{X}_t, t)$, which is used to estimate the clean image at time step 0 via the reverse diffusion formulation

$$\hat{\mathcal{X}}_0 = \frac{1}{\sqrt{\bar{\alpha}_t}}\left(\mathcal{X}_t - \sqrt{1 - \bar{\alpha}_t}\,\hat{\mathcal{E}}_\phi\right) \tag{6}$$

where $\hat{\mathcal{X}}_0$ represents the residual image after background suppression. This operation subtracts the predicted noise component from the noisy observation and rescales the result to the original signal magnitude. In the context of background suppression, $\hat{\mathcal{X}}_0$ naturally preserves structures that deviate from the learned background distribution.

B. Lightweight Fractional Fourier Transform

The proposed LFrFT-based spectral analysis framework is illustrated in Fig. 2. For the spectral vector $x \in \mathbb{R}^B$ of any pixel in the HSI, the spectral and frequency dimensions of the hyperspectral anomaly targets in different scenarios are significantly different. To enable the model to adaptively match the optimal analysis dimension in different scenarios, we introduce a learnable parameter θ ($\theta \in [0, 1]$). θ is a zero-dimensional single global scalar with no spectral, spatial, or sample dimensions, shared across the entire training dataset, and it is not learned per image, per pixel, or per spectral band. It achieves continuous rotation of the spectrum from the spectral domain to the frequency domain. The physical significance of this operation is dynamically switching the analytical dimensions of the spectrum. When $\theta = 1$, it degrades to the standard Fourier transform, that is, complete frequency-domain analysis. When $\theta = 0$, it degrades to the original spectrum, that is, complete spectral-domain analysis. When θ takes an intermediate value, it can simultaneously capture the spectral morphology in the spectral domain and the characteristics in the frequency domain. Its mathematical definition is

$$\mathcal{F}^\theta(x) = \sqrt{\frac{1 - j\cot\frac{\pi\theta}{2}}{2\pi}} \int_{-\infty}^{\infty} x(v) \exp\left(j\frac{u^2 + v^2}{2}\cot\frac{\pi\theta}{2} - juv\csc\frac{\pi\theta}{2}\right) dv \tag{7}$$

where $\mathcal{F}^\theta(\cdot)$ denotes the LFrFT operator of normalized fractional order θ, $j = \sqrt{-1}$ is the imaginary unit, and $\pi\theta/2$ is the corresponding rotation angle. Moreover, v denotes the input spectral-domain variable, whereas u denotes the output variable in the fractional Fourier domain. The cases $\theta = 0$ and $\theta = 1$ are understood in the limiting sense and correspond to the identity and standard Fourier transforms, respectively.

In practical implementation, to adapt to the parallel computing of graphics processing units (GPUs), we adopt the numerical approximation algorithm of LFrFT. First, we perform the standard fast Fourier transform (FFT) on the spectral vector x to obtain the frequency-domain representation $X(f)$

$$X(f) = \mathcal{F}\{x\} \tag{8}$$

where $\mathcal{F}\{\cdot\}$ denotes the 1-D FFT operator for spectral vector transformation. Then, the spectral–frequency rotation is achieved through learnable fractional phase modulation

$$X'(f) = X(f) \cdot \exp\left(-j\pi\theta f^2\right) \tag{9}$$

where $X'(f)$ denotes the spectral–frequency mixed feature after phase modulation and $\exp(-j\pi\theta f^2)$ is a phase slope controlled by θ, whose function is to apply differentiated phase offsets to different frequency components, thereby highlighting the frequency components related to anomalies.

To precisely screen and enhance the frequency components related to anomalies, we introduce a lightweight multilayer perceptron (MLP) composed of two fully connected layers to generate amplitude modulation weights. Concretely, this two-layer fully connected network takes the per-pixel B-dimensional spectral–frequency feature $\mathcal{F}^\theta(x)$ after the fractional Fourier transform as input. The first fully connected layer compresses the number of spectral bands down to half of the original count with a rectified linear unit (ReLU) activation function, and the second fully connected layer recovers the number of spectral bands back to the original quantity to output adaptive bandwise amplitude modulation weights. This lightweight design can adaptively learn the importance of different frequency components while avoiding excessive increase in computational load.

Specifically, the spectral–frequency mixed feature $X'(f) = \mathcal{F}^\theta(x)$ after the fractional Fourier transform is fed into the

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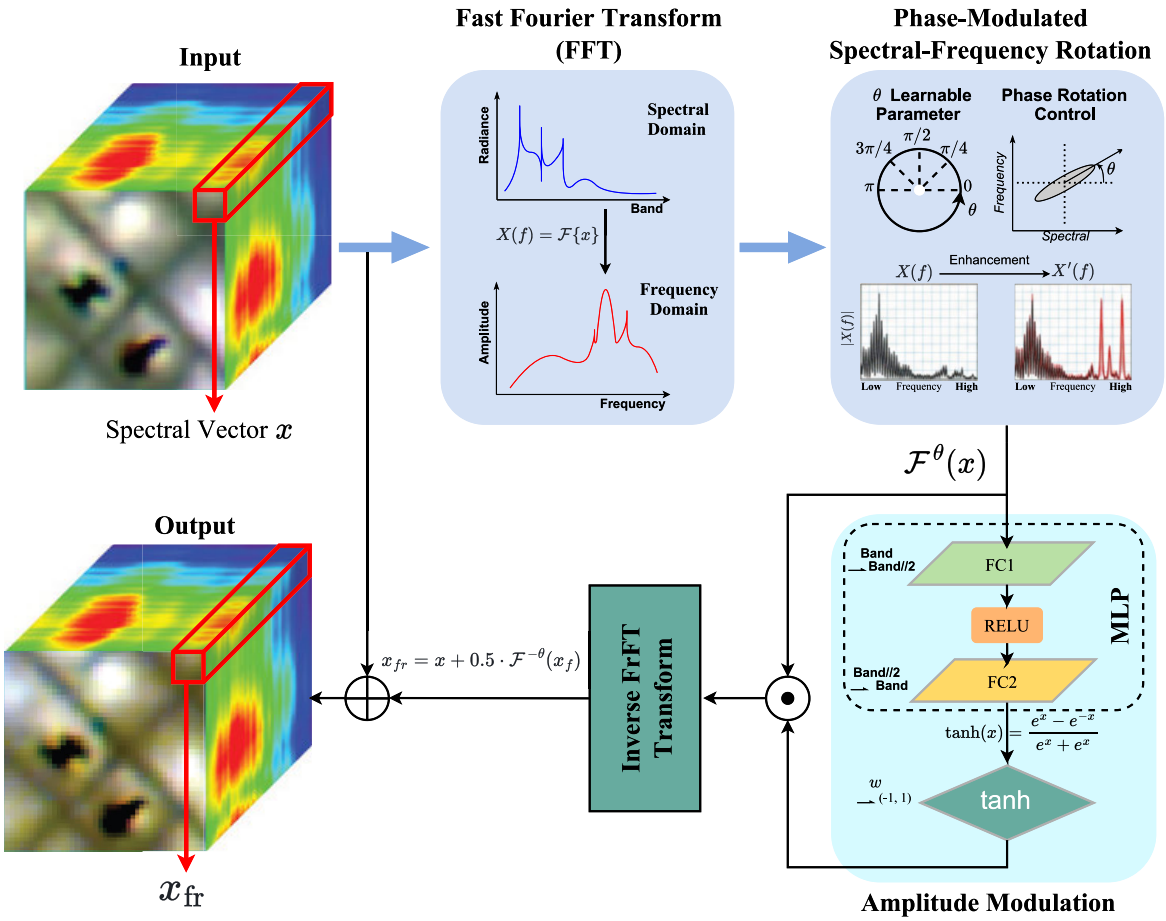


Fig. 2. Architecture of the proposed LFrFT module. The input spectral vector is transformed into the frequency domain via FFT, followed by phase modulation with a learnable fractional order θ and amplitude modulation using an MLP. The inverse LFrFT then maps the enhanced features back to the spectral domain, which are fused with the original spectrum through the residual connection.

aforementioned two-layer MLP for nonlinear dimension mapping to obtain the initial weight vector. The tanh activation function is connected to confine the weight range within $[-1, 1]$, and finally, the amplitude modulation weight w is generated

$$w = \tanh\left(\mathbf{FC}_2\left(\mathrm{ReLU}\left(\mathbf{FC}_1\left(\mathcal{F}^{\theta}(x)\right)\right)\right)\right) \tag{10}$$

where $\mathbf{FC}_1$ and $\mathbf{FC}_2$ denote the first and second fully connected layers of the MLP and ReLU$(\cdot)$ represents the rectified linear unit activation function. Among them, the core function of tanh activation is to prevent excessive modulation of spectral features caused by overly large weight values, while allowing the weights to be either positive or negative. By multiplying the generated amplitude modulation weight w elementwise with the frequency-domain feature $\mathcal{F}^{\theta}(x)$, the enhancement of abnormally correlated frequencies and the suppression of background redundant frequencies are achieved

$$\begin{aligned} x_{\mathrm{f}} &= w \odot \mathcal{F}^{\theta}(x) \\ &= \tanh\left(\mathbf{FC}_2\left(\mathrm{ReLU}\left(\mathbf{FC}_1\left(\mathcal{F}^{\theta}(x)\right)\right)\right)\right) \odot \mathcal{F}^{\theta}(x) \end{aligned} \tag{11}$$

where x_{f} denotes the frequency-domain feature after amplitude modulation and $\odot$ represents the element-by-element multiplication operation. To maintain the basic form of the original spectrum and avoid losing the key spectral profile through frequency-domain transformation, we fuse the original spectrum with the enhanced features after the inverse frequency-domain transformation through residual paths

$$x_{\mathrm{fr}} = x + 0.5 \cdot \mathcal{F}^{-\theta}(x_{\mathrm{f}}) \tag{12}$$

where x_{fr} denotes the final enhanced spectral vector after residual fusion and $\mathcal{F}^{-\theta}(\cdot)$ represents the inverse fractional Fourier transform. The parameters of the MLP and θ can be adaptively adjusted during model training and can automatically focus on the most distinguishable spectral–frequency dimension for hyperspectral data in different scenarios, ultimately enabling the enhanced spectral x_{fr} to more accurately depict the subtle spectral differences of weak and small targets. We next discuss whether this single zero-dimensional global transformation angle θ is sufficient for hyperspectral data with highly heterogeneous backgrounds. Although the fractional rotation operation adopts one unified global θ for all spectral vectors, the subsequent pixel-level MLP amplitude modulation module generates independent bandwise modulation weights w for each individual pixel to compensate for scene

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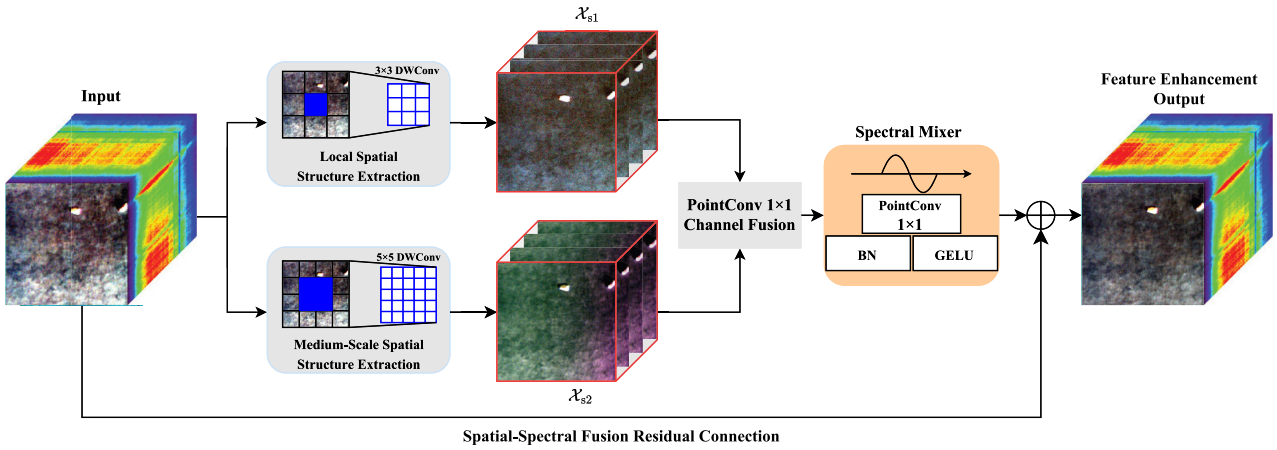


Fig. 3. Architecture of the proposed MSSE module. The input HSI is processed by 3×3 and 5×5 depthwise convolutions to capture local and medium-scale spatial features, which are fused through a 1×1 pointwise convolution. A spectral mixer is then employed to refine the fused representations, followed by a residual connection that integrates the enhanced features with the original input to generate the final output.

heterogeneity. The coupled design of global spectrum rotation and pixel-specific frequency amplitude adjustment jointly adapts to varying background and target spectral–frequency characteristics, eliminating the requirement for pixelwise or bandwise learnable rotation angles.

C. Multiscale Spatial–Spectral Enhancement

The proposed MSSE module is illustrated in Fig. 3. Spatial neighborhood context implies that the representation of a given pixel is correlated with those of its surrounding pixels, whereas anomaly targets typically exhibit spatial inconsistencies with the local background. In HSIs, such anomalies may appear at different spatial scales, ranging from isolated pointwise disturbances to mesoscale regional structures.

To accommodate these multiscale spatial characteristics and avoid the scale bias introduced by a single convolutional kernel, the MSSE module employs multiscale depthwise separable convolutions to capture fine-grained local structures and broader contextual information, respectively. Specifically, $\hat{\mathcal{X}}$ is the HSI cube processed by LFrFT. The local fine structures corresponding to pointwise or small-size anomalies are first extracted using a 3×3 depthwise separable convolution

$$\mathcal{X}_{s1} = \text{DepthwiseConv}_{3\times3}\left(\hat{\mathcal{X}}\right) \tag{13}$$

where $\text{DepthwiseConv}_{3\times3}(\cdot)$ denotes a 3×3 depthwise separable convolution operation that decomposes standard convolution into depthwise convolution and pointwise convolution. The depthwise convolution step convolves each spectral band independently with a 3×3 kernel, while the pointwise convolution step fuses spectral information via 1×1 convolution.

Then, the mesoscale context associations that adapt to small regional anomalies are captured through 5×5 depth-separable volume accumulation

$$\mathcal{X}_{s2} = \text{DepthwiseConv}_{5\times5}\left(\hat{\mathcal{X}}\right) \tag{14}$$

where $\text{DepthwiseConv}_{5\times5}$ denotes a 5×5 depthwise separable convolution operation for capturing mesoscale spatial context associations.

Depth-separable convolution decouples spatial convolution from channel fusion. While maintaining the spatial feature extraction capability, the number of parameters is only $1/B$ of that of ordinary convolution, effectively avoiding the computational redundancy caused by high channels in hyperspectral data and ensuring the efficiency and effectiveness of multiscale feature extraction. We achieve deep fusion of spatial spectral features through lightweight convolution operations. First, channel stitching is performed on multiscale spatial features $\mathcal{X}_{s1}$ and $\mathcal{X}_{s2}$. Then, channel fusion of spatial features is completed through a 1×1 point convolution, compressing redundant dimensions while integrating spatial information of different scales

$$\mathcal{X}_{sf} = \text{PointConv}_{1\times1}\left(\text{Concat}\left(\mathcal{X}_{s1}, \mathcal{X}_{s2}\right)\right) \tag{15}$$

where $\mathcal{X}_{sf}$ denotes the fused spatial–spectral feature cube, $\text{PointConv}_{1\times1}$ represents a 1×1 pointwise convolution operation for channel fusion and dimension compression, and Concat denotes the channelwise concatenation operation of multiscale spatial features.

On this basis, a 1×1 point convolution in the spectral direction is introduced. Under the premise that the spatial dimension remains unchanged, the spectral vector at each spatial position is transformed through channels to achieve cross-dimensional mixing of spatial structure information and spectral difference information, yielding the enhanced spatiospectral feature $\mathcal{Y}$

$$\mathcal{Y} = \text{PointConv}_{1\times1}\left(\text{GELU}\left(\text{BN}\left(\text{PointConv}_{1\times1}\left(\mathcal{X}_{sf}\right)\right)\right)\right). \tag{16}$$

The introduction of batch normalization (BN) and Gaussian error linear unit (GELU) activation can alleviate the training instability problem caused by the distribution fluctuations of hyperspectral data, making the features after spatial–spectral fusion more discriminative.

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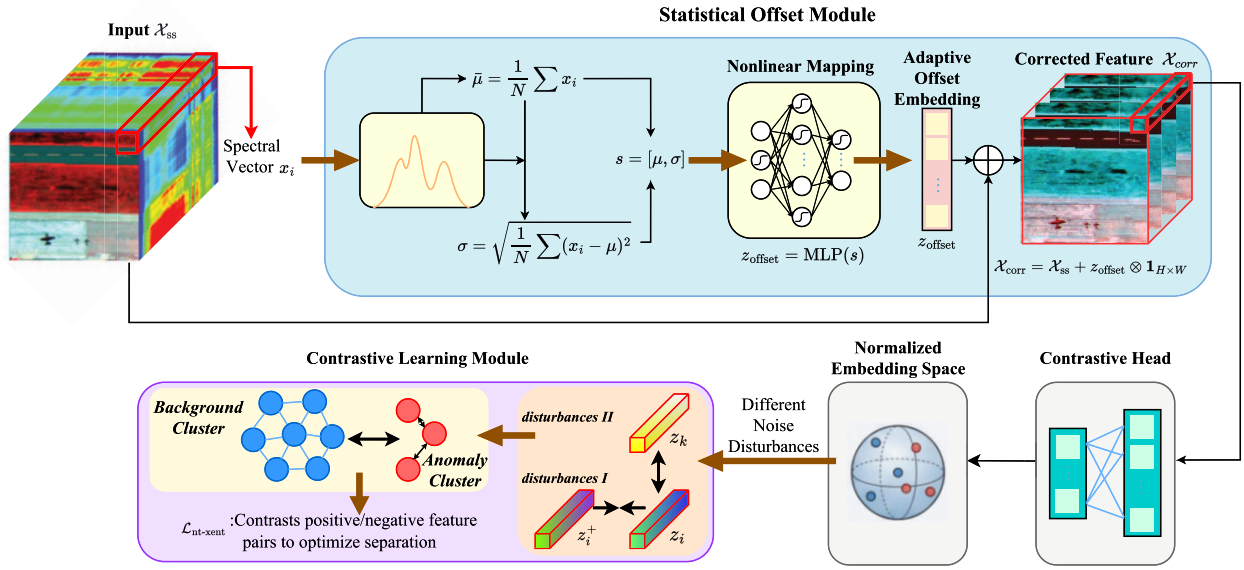


Fig. 4. Architecture of the proposed SACL module. The module first computes the spectral mean and standard deviation to construct a statistical descriptor, which is mapped to an adaptive offset embedding via an MLP for feature distribution correction.

To avoid the loss of the original spectral basic information caused by directly replacing the original input with the fused empty spectral features, and to prevent the model from being abnormally insensitive to significant spectral differences but weak spatial features, we adopt a residual connection to embed the enhanced features into the backbone network, achieving the parallel flow of the original features and the enhanced features

$$\mathcal{X}_{ss} = \mathcal{X} + \mathcal{Y}. \tag{17}$$

The final output $\mathcal{X}_{ss}$ simultaneously possesses spectral discrimination and spatial consistency.

D. Statistical Adaptive and Contrastive Learning

The proposed SACL module is illustrated in Fig. 4. We first calculate the mean $\mu \in \mathbb{R}^B$ and standard deviation $\sigma \in \mathbb{R}^B$ of the input HSI feature $\mathcal{X}_{ss} \in \mathbb{R}^{H\times W\times B}$ per spectral band across all spatial pixels, construct a statistical vector $s = [\mu, \sigma] \in \mathbb{R}^{2B}$ by concatenating the bandwise mean and standard deviation, and then generate an adaptive offset embedding $z_{offset} \in \mathbb{R}^B$ that matches the spectral feature dimension through nonlinear mapping

$$z_{offset} = \mathrm{MLP}(s). \tag{18}$$

The learnable parameters of the MLP can adaptively adjust the mapping relationship during training, enabling the offset embedding to better match the background distribution characteristics in different scenarios. Subsequently, the embedding dynamically corrects the feature distribution by adding elements one by one, yielding the distribution-corrected feature $\mathcal{X}_{corr}$

$$\mathcal{X}_{corr} = \mathcal{X}_{ss} + z_{offset} \otimes \mathbf{1}_{H\times W} \tag{19}$$

where $\mathbf{1}_{H\times W}$ denotes an $H \times W$ full 1-matrix consistent with the spatial dimensions of $\mathcal{X}_{ss}$ and $\otimes$ denotes the outer product operation that broadcasts the B-dimensional offset vector to form an $H \times W \times B$ spatial–spectral feature cube matching the dimension of $\mathcal{X}_{ss}$. $\mathcal{X}_{corr}$ is sent to the denoising network $f_\phi(\mathcal{X}_t, t)$, where the depth encoder composed of residual layers is used. Combined with the time-step embedding information of the diffusion model, the depth encoding of features is completed. Finally, the background noise estimation $\hat{\mathcal{E}} \in \mathbb{R}^{H\times W\times B}$ consistent with the dimensions of the original HSI is generated through the output layer.

To alleviate feature overlap and insufficient semantic discrimination between weak anomaly features and background features in high-dimensional spaces, the corrected representations are projected into a normalized embedding space. An unsupervised contrastive learning strategy is adopted, which does not require prior partitioning of background and anomaly samples. Instead, two perspective views with different noise perturbations are generated for the same HSI pixel. Let the resulting feature pair be denoted as z_i and z_i^+, and the contrastive loss is defined as

$$\mathcal{L}_{nt\text{-}xent} = -\frac{1}{2N}\sum_{i=1}^{2N} \log \frac{\exp\left(\mathrm{sim}\left(z_i, z_i^+\right)/\tau\right)}{\sum_{\substack{k=1\\k\neq i}}^{2N} \exp\left(\mathrm{sim}\left(z_i, z_k\right)/\tau\right)} \tag{20}$$

where N denotes the number of pixels in a training batch, $\mathrm{sim}(\cdot)$ denotes the cosine similarity, and τ is the temperature parameter, which is set to 0.1 in our experiments. For each anchor embedding z_i, z_i^+ denotes its positive counterpart generated from another noise-perturbed view of the same pixel. The denominator contains all candidate embeddings except the anchor itself, including its positive counterpart and the negative embeddings generated from different pixels.

In actual training, this method adopts a dual-loss collaborative optimization strategy. The abovementioned contrastive learning loss $\mathcal{L}_{nt\text{-}xent}$ and the denoising loss $\mathcal{L}_{denoise}$ are weighted and combined into the total loss, which jointly drives the update of network parameters. The total loss formula is

$$\mathcal{L}_{total} = \mathcal{L}_{denoise} + \lambda \cdot \mathcal{L}_{nt\text{-}xent} \tag{21}$$

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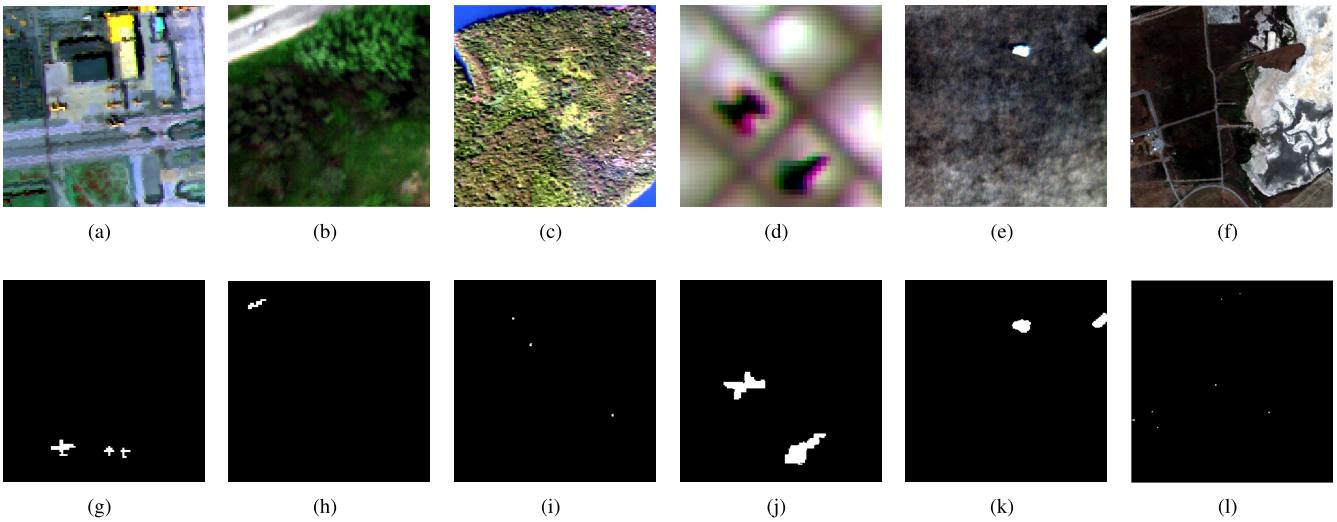


Fig. 5. Pseudocolor image and ground-truth map of the six hyperspectral datasets. (a) Gulfport. (b) Luoyang Vehicle. (c) Countryside. (d) SAU Aircraft. (e) BIT Rock. (f) Wasteland. (g) Gulfport (ground truth). (h) Luoyang Vehicle (ground truth). (i) Countryside (ground truth). (j) SAU Aircraft (ground truth). (k) BIT Rock (ground truth). (l) Wasteland (ground truth).

where $\mathcal{L}_{\text{denoise}}$ is the core L_2 loss of the denoising network and λ is the weight of the contrastive loss. During training, the gradient of the total loss is backpropagated to all learnable parameters of the network.

This module adopts a collaborative design that integrates statistical adaptive distribution correction with contrastive learning-based semantic discrimination. By mitigating feature distribution heterogeneity in complex backgrounds, it enhances the discriminability of weak anomalies and maintains stable detection performance under high background noise.

TABLE I
DETAILED PARAMETER SETTINGS OF THE PROPOSED FDCRN

Module/Setting	Config	Value
Denoising Network	Hidden dims	[200, 100, 50]
	Pre-embed dim	128
Diffusion Params	Total steps (T)	1000
	Inference steps (t)	500
Training	Epochs	1000
	Learning rate	10^{-6}
Enhancement	Residual repeats (K)	20
	Contrastive weight (λ)	1.0

III. EXPERIMENTS AND ANALYSIS

A. Datasets and Experimental Setup

1) Datasets: The pseudocolor images and ground-truth maps of the six hyperspectral datasets are shown in Fig. 5. The first dataset is Gulfport, captured by the Rosis sensor over an airport apron area, with a spatial resolution of 100×100 pixels and 205 bands. The second dataset is Luoyang Vehicle, collected by the GaiaSky-mini2 dualix spectral imaging hyperspectral sensor in Luoyang, Henan Province. This scene is dominated by dense vegetation, with vehicles set as anomalous targets against the tree background, and it has a spatial size of 100×100 pixels with 176 spectral bands. The third dataset, referred to as Countryside, was acquired over a countryside scene, where the anomalies correspond to small-scale military camouflaged targets. This dataset has a spatial size of 200×200 pixels and 48 bands. The fourth dataset is SAU Aircraft, acquired by the GaiaSky-mini2 dual-channel hyperspectral sensor at Shenyang Aerospace University. This scene covers an airport pavement area, with parked aircraft defined as anomalous targets, and it has a spatial size of 200×200 pixels with 126 spectral bands. The fifth dataset is BIT Rock, collected by the GaiaSky-mini2 dual-channel hyperspectral sensor on the campus of Beijing Institute of Technology. This dataset covers bare soil terrain with scattered small rock patches treated as anomalies, and it contains 256×256 pixels with 176 spectral bands. The sixth dataset, referred to as Wasteland, was captured over a wasteland area, consisting of 512×512 pixels and 198 bands, where the anomalous objects are tiny-sized camouflaged military vehicles distributed across the whole scene.

2) Parameter Configuration: The parameter configuration covers denoising network architecture, diffusion process settings, training hyperparameters, and enhancement module adjustments. Detailed values are summarized in Table I. For the denoising network, a three-layer feedforward structure with decreasing hidden dimensions is designed to capture hierarchical features, paired with a 128-D preembedding layer for effective input representation. Regarding diffusion parameters, 1000 total forward diffusion steps (T) are used to gradually inject noise, while 500 inference steps (t) are selected to balance generation quality and speed during reverse denoising. In practical applications for large-scale hyperspectral data, the number of inference steps t can be appropriately reduced. Experimental results demonstrate that the model maintains stable anomaly detection accuracy when adopting fewer inference steps, such as 100 and 200, which effectively alleviates the heavy computational overhead caused by excessive iteration steps on large-size input cubes. For training, the model is trained for 1000 epochs with a learning rate of 10^{-6} to ensure stable convergence. In the enhancement module, 20 residual

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repeats (K) strengthen feature propagation, and a contrastive weight (λ) of 1.0 balances denoising loss and contrastive learning loss.

3) Comparative Methods: To verify the effectiveness of FDCRN in complex scenarios, seven methods are selected for comparison, covering traditional statistical modeling, kernel-based isolation, and advanced deep learning approaches. The classic RX algorithm [33] is a representative statistical-based method that detects anomalies by measuring the spectral deviation of each pixel from the global background distribution. KIFD [42] enhances anomaly–background separability via kernel mapping and isolation forest, optimizing spatial information via local window iteration. TAEF [53] captures long-range spatiospectral dependencies via Transformer attention and detects anomalies via reconstruction errors. HFGDM [52] constructs a frequency-aware diffusion framework to decouple high-frequency anomaly features and low-frequency background components for progressive background suppression. FERD [23] leverages reverse distillation feature enhancement to excavate subtle anomalous differences and improve discrimination for weak targets. MrsNet [54] adopts a multiscale dual-domain masked reconstruction paradigm to constrain consistent background representation and suppress false alarms. BSDM [55] exploits pseudobackground noise to learn background distribution, improving diffusion inference for background suppression.

4) Evaluation Metrics: The receiver operating characteristic (ROC) curve characterizes the tradeoff in detection performance under varying decision thresholds, where the false positive (FP) rate is plotted along the horizontal axis and the true positive (TP) rate along the vertical axis. To compute these two rates quantitatively, four fundamental quantities are defined: TP, representing anomaly pixels correctly detected as anomalies; false negative (FN), denoting anomaly pixels incorrectly classified as background; true negative (TN), corresponding to background pixels correctly classified as background; and FP, indicating background pixels incorrectly classified as anomalies.

Based on the above definitions, the TP rate P_d and the FP rate P_f are defined as

$$P_d = \frac{TP}{TP + FN} \tag{22}$$

$$P_f = \frac{FP}{TN + FP}. \tag{23}$$

The area under the ROC curve (AUC) is defined as the area enclosed by the ROC curve and the horizontal axis. In this study, the AUC is numerically approximated by uniformly sampling p decision thresholds $\delta \in [0, 1]$ (with $p = 1000$) and accumulating the areas of adjacent trapezoids along the ROC curve

$$\text{AUC} = \frac{1}{2} \sum_{m=1}^{p-1} \left(P_d^{(m+1)} + P_d^{(m)} \right) \cdot \left(P_f^{(m+1)} - P_f^{(m)} \right) \tag{24}$$

where $P_d^{(m)}$ and $P_f^{(m)}$ denote the TP rate and FP rate corresponding to the mth threshold δ_m, respectively. The value range of AUC is $[0, 1.0]$. A value closer to 1 indicates that the algorithm can achieve higher P_d with lower P_f, which is the core quantitative indicator to measure the overall detection accuracy.

The box–whisker plot (box plot) is employed to assess the separability between anomaly and background pixels by comparing their detection response distributions. In the box plot, the green and red boxes correspond to background and anomaly pixels, respectively, where the box boundaries indicate the 25th (Q1) and 75th (Q3) percentiles, the central line denotes the median, and the whiskers extend to values within 1.5 times the interquartile range (IQR = Q3− Q1). A lower and more compact background box indicates more effective background suppression, while a higher-positioned anomaly box reflects an increased response gap between anomalies and background, demonstrating stronger anomaly discrimination capability.

The runtime is measured as the total time required for a complete detection process (in seconds), including data preprocessing, model inference (or algorithm computation), and result post-processing, which reflects the practical efficiency of the method. All methods are evaluated under the same hardware and software environment, and the reported runtime is averaged over three independent runs to reduce randomness.

The experiment is conducted on a workstation equipped with an NVIDIA RTX 5060 GPU, an Intel Core i7-14650HX CPU, 16 GB of RAM, and a 1-TB PC SN500S WD solid-state drive. This hardware configuration ensures efficient loading of hyperspectral data and parallel training of the model. The operating system is Windows 11 24H2. The deep learning framework adopts PyTorch 2.8.0, combined with CUDA 12.8 for GPU-accelerated computing. The data preprocessing and visualization tools include NumPy 1.26.4, OpenCV 4.12.0, and Matplotlib 3.10.7.

B. Comparative Results and Analysis

1) Visual Effect Comparison: Fig. 6 presents the grayscale detection results of FDCRN and the seven comparative methods on the six hyperspectral datasets, where brighter pixels indicate higher anomaly confidence. From the visual comparisons, FDCRN demonstrates superior performance in three main aspects. First, FDCRN achieves clear anomaly highlighting with low background clutter. On the Gulfport airport scene, airport anomaly targets are accurately detected as continuous bright regions, whereas RX generates scattered background noise near targets, KIFD introduces obvious granular clutter interference, TAEF fails to completely suppress background fluctuation, and HFGDM produces faint false responses in pavement background areas. This improvement can be attributed to the LFrFT module, which enhances high-frequency spectral fluctuations of anomalous targets, together with the MSSE module that effectively suppresses interference from complex pavement background. Second, FDCRN exhibits strong capability in detecting weak and small targets. In the Countryside containing small-scale camouflaged military objects, most comparative methods show degraded performance: RX and FERD yield blurred and fragmented target responses, KIFD confuses subtle target signatures with surrounding farmland backgrounds, and BSDM detects only incomplete target pixels. In contrast, FDCRN consistently

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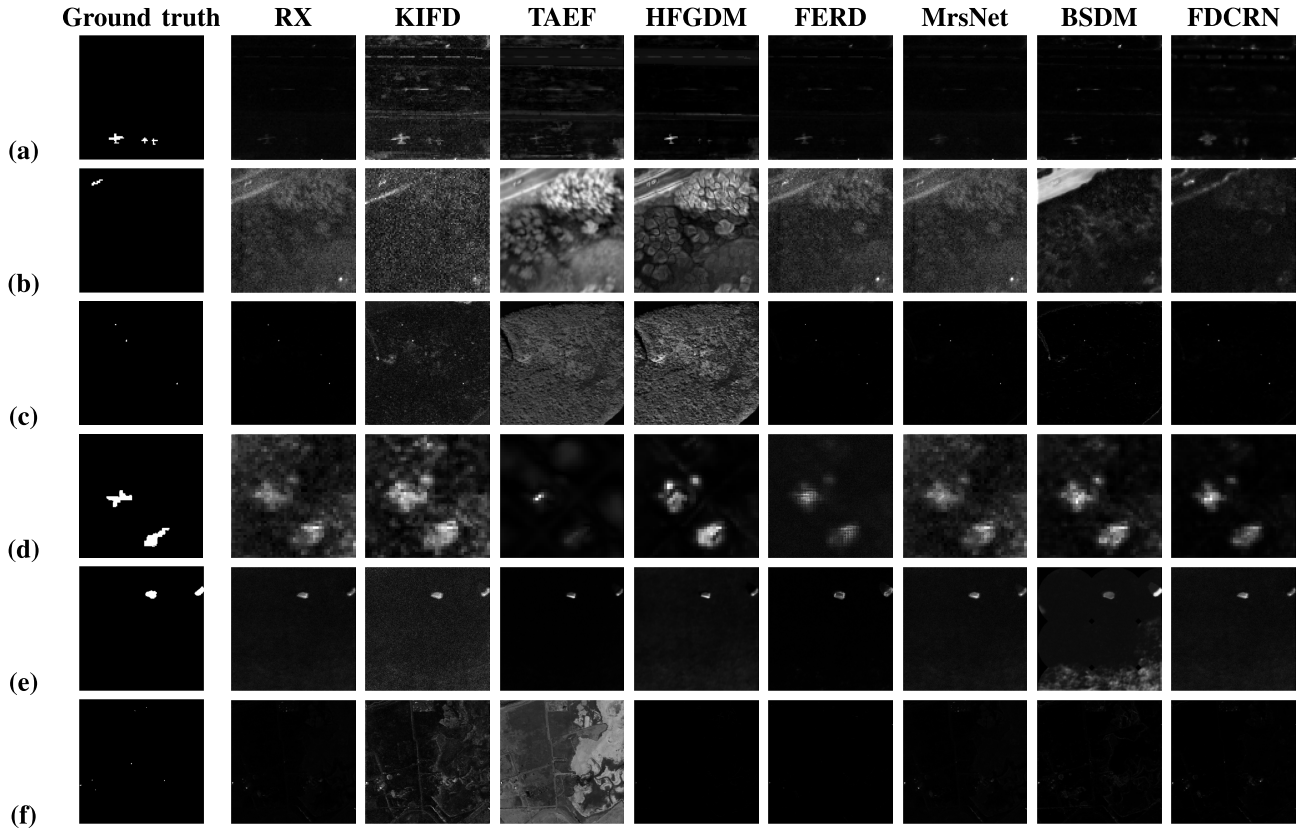


Fig. 6. Ground truth and detection results of FDCRN and comparative methods on the six datasets. (a) Gulfport, (b) Luoyang Vehicle, (c) Countryside, (d) SAU Aircraft, (e) BIT Rock, and (f) Wasteland.

TABLE II

AUC VALUES FOR THE COMPARED APPROACHES ON THE SIX DATASETS. BEST RESULTS ARE IN BOLDFACE

Method	Gulfport	Luoyang Vehicle	Countryside	SAU Aircraft	BIT Rock	Wasteland
RX	0.9526	0.9480	0.8877	0.9442	0.9767	0.6358
KIFD	0.9676	0.9424	0.9029	0.9715	0.9788	0.6352
TAEF	0.8272	0.8206	0.6641	0.9723	0.9901	0.7501
HFGDM	0.9410	0.8702	0.6743	0.9810	0.9341	0.8988
FERD	0.9849	0.9565	0.9813	0.9815	0.9770	0.9700
MrsNet	0.9660	0.9481	0.9214	0.9619	0.9922	0.5137
BSDM	0.9783	0.9749	0.8288	0.9833	0.9749	0.7568
FDCRN	**0.9911**	**0.9983**	**0.9982**	**0.9897**	**0.9997**	**0.9909**

identifies all camouflaged targets as compact bright regions without obvious missed detections or false alarms, benefiting from the joint effects of fractional-domain feature enhancement and statistical distribution correction. Third, FDCRN shows robust adaptability to complex large-scale heterogeneous scenes. On the Wasteland with tiny camouflaged military vehicle anomalies, baseline methods such as RX and TAEF suffer from severe widespread background noise, while HFGDM and MrsNet partially suppress background interference but fail to form complete, sharp target contours. FDCRN accurately localizes camouflaged vehicles as salient concentrated bright regions against the complex wasteland background, indicating that the MSSE module effectively aggregates long-range multiscale contextual information, and the diffusion-driven background suppression framework provides stable elimination of complicated background clutter.

2) Quantitative Comparison: Table II reports the AUC values of FDCRN and the seven comparative methods on the six datasets, where AUC serves as a key metric for overall detection performance, ranging from 0 to 1.0. FDCRN achieves the highest AUC on all six datasets, demonstrating consistent and comprehensive superiority over RX, KIFD, TAEF, HFGDM, FERD, MrsNet, and BSDM. Its performance advantage is more prominent on datasets containing weak tiny targets and heterogeneous complex backgrounds, including Luoyang Vehicle, Countryside, and Wasteland, while, on

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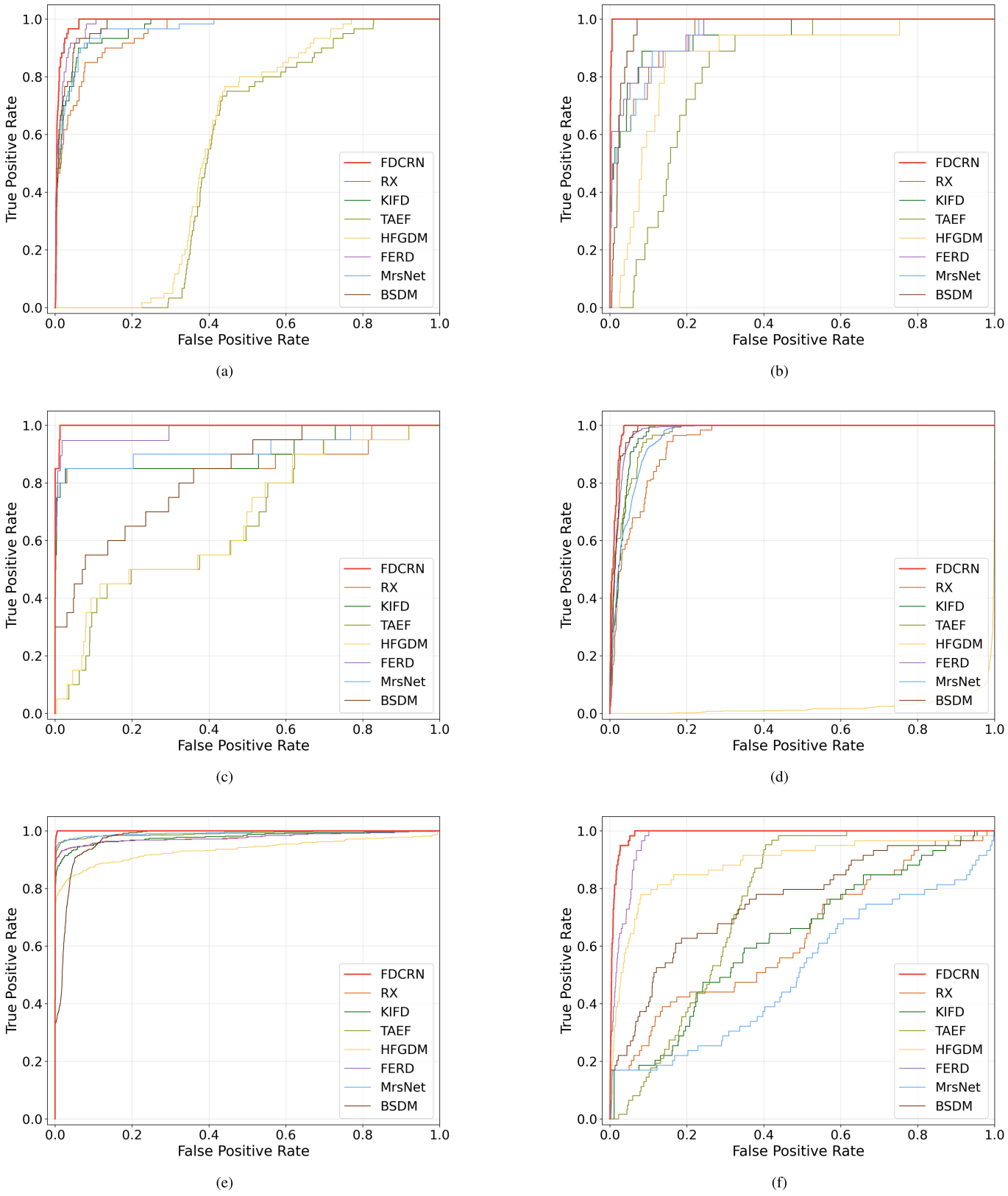


Fig. 7. ROC curves of the compared methods on the six hyperspectral datasets. (a) Gulfport, (b) Luoyang Vehicle, (c) Countryside, (d) SAU Aircraft, (e) BIT Rock, and (f) Wasteland.

relatively simple scenes, such as BIT Rock and SAU Aircraft, it still retains a clear leading margin against all competitors. All AUC values of FDCRN exceed 0.9890, with five out of six datasets reaching values above 0.9900. Among all competitors, only FERD delivers competitive overall performance, yet it still exhibits obvious performance degradation on Wasteland and Gulfport compared to FDCRN. Traditional statistical RX and KIFD suffer severe performance drops on the

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large-scale Wasteland scene, and reconstruction-oriented TAEF and MrsNet display unstable generalization across different scenes. The diffusion-based HFGDM and baseline BSDM are constrained by insufficient frequency-domain modeling and incomplete spatiospectral interaction, failing to match the detection accuracy of the proposed FDCRN. These quantitative results fully verify the stable and reliable detection capability of our method across diverse scene types, target sizes, and background complexities.

3) ROC Curve Analysis: From the ROC curve distributions in Fig. 7, three key observations can be drawn. First, FDCRN achieves superior overall performance across all six datasets, with its ROC curves consistently located closest to the upper-left corner. Across diverse scenes including the Gulfport airport apron, Luoyang Vehicle vegetation-background scenario, Countryside camouflaged small targets, SAU Aircraft apron, BIT Rock bare-soil rocky anomalies, and large-scale Wasteland, FDCRN attains higher TP rates under identical FP rates against RX, KIFD, TAEF, HFGDM, FERD, MrsNet, and BSDM, delivering a more optimal balance between anomaly recall and background false-alarm suppression. Second, FDCRN exhibits strong adaptability to complex large-scale scenes, particularly on the Wasteland with tiny camouflaged military vehicle anomalies. While most comparative methods present slowly rising curves and limited final detection accuracy, FDCRN rises sharply within the low false-positive region and quickly approaches near-perfect detection rates. This advantage originates from the LFrFT module for excavating subtle frequency-domain differences between targets and backgrounds, as well as the MSSE module that aggregates long-range spatial contextual cues comprehensively. Third, FDCRN maintains steady detection performance on every evaluated dataset, whereas several competitors suffer obvious performance decline on specific challenging scenes. Such favorable cross-scene generalization mainly benefits from the statistical distribution calibration function embedded in the SACL module.

Overall, the ROC curve results demonstrate that FDCRN consistently achieves a more effective tradeoff between anomaly detection and background suppression across different scene types, spatial scales, and target characteristics.

4) Visualization of Separation Performance: The box plot results in Fig. 8 demonstrate that FDCRN consistently achieves superior anomaly–background separation across all six datasets. Specifically, the background response distribution of FDCRN is tightly concentrated near zero with a narrow value range, realizing more thorough background clutter suppression against the seven competitors. This favorable property originates from the diffusion-driven background suppression backbone paired with the distribution calibration effect of the proposed SACL module. Meanwhile, FDCRN yields markedly higher median and upper quantile values for anomaly responses than all alternative approaches, verifying that the LFrFT fractional frequency enhancement module and multiscale MSSE spatiospectral aggregation module effectively strengthen the salience of weak anomalous signatures. Furthermore, FDCRN creates the clearest statistical gap between anomaly and background populations with negligible distribution overlap, even for difficult heterogeneous scenarios such as the large-scale Wasteland with tiny camouflaged vehicle targets. In contrast, most competing methods present obvious overlapping regions between anomaly and background response distributions, leading to ambiguous discrimination boundaries. These quantitative distribution comparisons comprehensively validate that FDCRN is capable of restraining background fluctuations adequately while intensifying the discriminative response of subtle anomalies, possessing outstanding robustness for complex and diverse hyperspectral scenes.

5) Running Time Comparison: As illustrated in Table III, the proposed FDCRN achieves a favorable balance between detection performance and computational efficiency. The traditional statistical RX and kernel-based KIFD possess the shortest running time across all datasets, but they yield unsatisfactory detection accuracy and poor anomaly–background discrimination ability for complex heterogeneous scenes. Deep learning-based competitors including TAEF, HFGDM, FERD, MrsNet, and BSDM generally introduce heavier computational overhead. Specifically, Transformer-structured TAEF, reconstruction-oriented MrsNet, and distillation-based FERD bring steadily rising time consumption as image spatial size increases, while diffusion-based HFGDM and BSDM also suffer from nonnegligible inference latency caused by iterative generation processes. In contrast, FDCRN generally maintains competitive inference efficiency compared with the deep learning-based methods. This efficiency benefit originates from the lightweight architecture design of the LFrFT, MSSE, and SACL core modules, which eliminates redundant feature computation while preserving outstanding detection precision. Even when processing the largest-size Wasteland hyperspectral scene, FDCRN achieves obvious runtime superiority against most deep learning competitors, verifying its satisfactory engineering practicability and deployment potential for real-time HAD tasks.

C. Ablation Experiments

To validate the necessity and synergistic effect of each core module in the proposed FDCRN, we design systematic ablation experiments and conduct quantitative/qualitative analysis. The following details the experiment design and in-depth discussion of module effectiveness.

1) Design of Ablation Experiments: The core idea of ablation experiments is to sequentially remove one or a group of core modules while keeping other configurations consistent with the original FDCRN. This ensures that performance differences between model variants are solely attributed to the removed modules, avoiding interference from irrelevant factors.

Based on the three key innovations of FDCRN, four model variants are constructed. FDCRN is the complete model integrating all core modules, including LFrFT, MSSE, and SACL projection head. FDCRN without LFrFT removes the spectral LFrFT module. The model directly uses raw hyperspectral spectral data for feature extraction, without adaptive frequency-domain modulation and residual fusion. FDCRN without MSSE cancels the MSSE module. Only a

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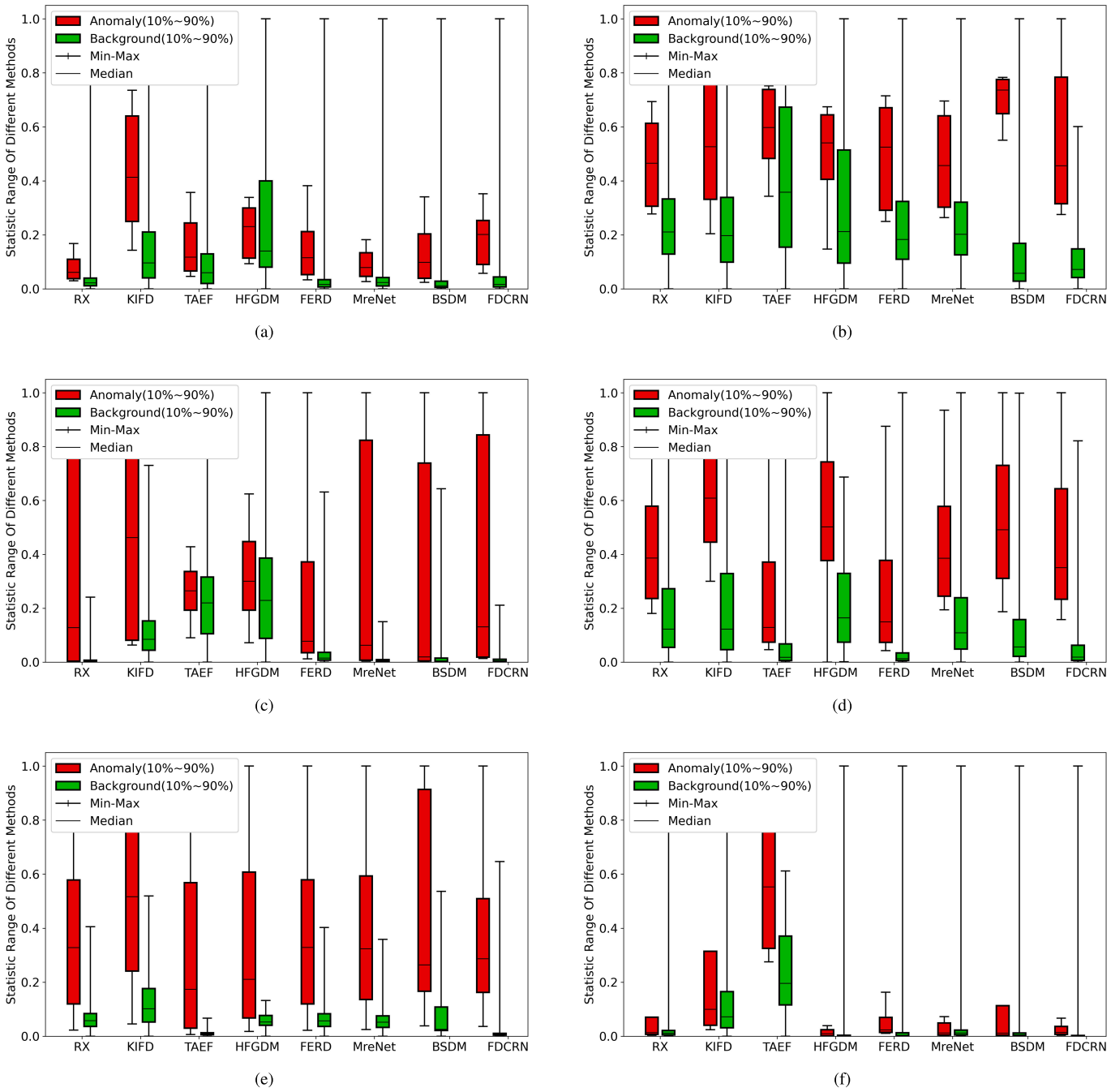


Fig. 8. Box plots of the compared approaches on the six datasets. (a) Gulfport, (b) Luoyang Vehicle, (c) Countryside, (d) SAU Aircraft, (e) BIT Rock, and (f) Wasteland.

single 3×3 standard convolution is used for spatial feature extraction, abandoning multiscale context aggregation and depth-separable convolution. FDCRN without SACL removes the SACL projection head. The model uses only the ℓ_2 denoising loss for training, without dynamic feature distribution correction and semantic separation of background/anomalies.

2) Effectiveness Analysis of Core Modules: The quantitative results of all ablation variants on the six hyperspectral datasets are shown in Table IV. The FDCRN model demonstrates significant performance improvements benefiting from the synergistic effect of its three core modules. Removing the LFrFT module causes obvious AUC degradation across all datasets, especially on Countryside, where the AUC drops from 0.9982 to 0.9747, which proves that LFrFT effectively excavates subtle frequency-domain spectral differences for weakly camouflaged targets. The MSSE module brings the most remarkable performance gain, and its removal leads to a severe AUC decline from 0.9909 to 0.9134 on the complex Wasteland scene, verifying that multiscale spatiospectral aggregation is critical for adapting to large-scale heterogeneous backgrounds and variable target sizes. Excluding the SACL module also yields a universal performance drop, with the most obvious decrement appearing on Countryside where AUC falls from 0.9982 to 0.9618, since SACL calibrates imbalanced background distribution and boosts discriminability between anomaly and background features via contrastive

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TABLE III
COMPLETE RUNNING TIME (SECONDS) OF THE COMPARED APPROACHES ON THE SIX DATASETS

Method	Gulfport	Luoyang Vehicle	Countryside	SAU Aircraft	BIT Rock	Wasteland
RX	1.51	2.72	1.36	2.54	2.51	3.15
KIFD	1.68	1.92	2.03	3.17	2.53	4.52
TAEF	74.08	53.52	102.86	189.89	305.87	480.62
HFGDM	58.25	83.23	88.56	74.08	102.86	480.62
FERD	151.76	158.69	45.59	375.84	568.24	876.01
MrsNet	41.73	44.47	62.67	151.08	179.30	701.48
BSDM	25.58	44.07	200.22	85.99	258.49	396.03
FDCRN	**15.47**	**35.60**	**58.89**	**76.67**	**119.71**	**276.93**

TABLE IV
AUC VALUES OF ABLATION VARIANTS ON THE SIX DATASETS. BEST RESULTS ARE IN BOLDFACE

Variant	Gulfport	Luoyang Vehicle	Countryside	SAU Aircraft	BIT Rock	Wasteland
w/o LFrFT	0.9901	0.9934	0.9747	0.9712	0.9853	0.9842
w/o MSSE	0.9911	0.9920	0.9890	0.9679	0.9955	0.9134
w/o SACL	0.9897	0.9962	0.9618	0.9759	0.9902	0.9889
FDCRN	**0.9953**	**0.9983**	**0.9982**	**0.9897**	**0.9997**	**0.9909**

projection. The complete FDCRN obtains the optimal AUC on all six datasets, which validates that LFrFT, MSSE, and SACL cooperate mutually: LFrFT optimizes spectral frequency characteristics, MSSE enriches multiscale feature representation, and SACL strengthens feature separability, forming a coordinated optimization pipeline with obvious complementary promotion effect.

IV. CONCLUSION

This study proposed a novel FDCRN framework for HAD. First, an LFrFT module was designed to adaptively modulate spectral frequency components through learnable fractional-order parameters, enabling effective capture of subtle frequency-domain characteristics of weak anomalies. Second, an MSSE was introduced to extract multiscale spatial contextual information via depthwise separable convolutions and to fuse spectral features, thereby strengthening spatial–spectral correlations of anomalous targets. Third, SACL combined with a contrastive learning-based projection head was employed to dynamically alleviate feature distribution shifts and improve semantic separability between background and anomalies across different scenes. Extensive experiments conducted on six real hyperspectral datasets demonstrated the effectiveness and robustness of the proposed FDCRN. The proposed method consistently achieved the highest AUC values, exceeding 0.9700 on all evaluated datasets, and outperformed seven state-of-the-art comparison methods in terms of weak target detection accuracy and complex scene generalization performance, while maintaining competitive computational efficiency. Future work will explore the integration of FDCRN with hyperspectral super-resolution techniques to enable subpixel anomaly detection, as well as the extension of the proposed framework to multitemporal hyperspectral data analysis.

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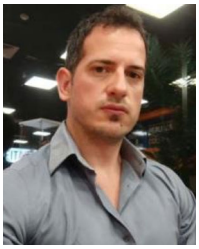
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